Name:J.Vishnu Vardhan
Reg No:192324297

ANNEXURE A
COMPARATIVE ANALYSIS

1: CNNs vs. Vision Transformers (ViTs) for Object Detection

Scenario: A self-driving car company needs to detect rare urban obstacles such as traffic cones and animals in real-time using limited labelled datasets.

1: Comparison of CNNs and Vision Transformers

Question: Compare CNNs and Vision Transformers in terms of feature extraction capabilities for local vs. global context, performance with small or sparse objects, and computational complexity and inference speed for embedded systems.

A. Feature Extraction – Local vs. Global Context

CNNs – Answer: Convolutional Neural Networks use convolutional filters that learn local patterns such as edges, corners, textures and object parts. Deeper layers combine these local features to form higher-level representations.

ViTs – Answer: Vision Transformers divide an image into patches and use self-attention to learn relationships between distant image regions. This gives them strong global-context modelling capability.

Comparison: CNNs are naturally strong at local feature extraction, while ViTs are particularly strong at modelling long-range relationships and global scene context. Modern CNNs can also capture broader context through deeper layers and multi-scale feature pyramids.

B. Performance with Small or Sparse Objects

CNNs – Answer: CNN-based detectors can use multi-scale feature maps and feature pyramids to detect objects at different sizes. This makes them effective for small obstacles such as distant traffic cones or animals.

ViTs – Answer: ViTs can use attention to identify relationships between an obstacle and surrounding scene regions. However, very small objects can be challenging if patch sizes are too large or if high-resolution information is lost.

Comparison: For small and sparse objects, a well-designed multi-scale CNN detector is often easier to optimize. Transformer-based detectors can also perform very well when sufficient resolution, pretraining and computational resources are available.

C. Computational Complexity and Inference Speed

CNNs – Answer: CNNs use highly optimized convolution operations and many lightweight architectures are available for embedded devices. This makes them suitable for real-time inference.

ViTs – Answer: Vision Transformers use self-attention, which can require substantial computation and memory, especially for high-resolution images with many patches. Efficient transformer variants reduce this cost.

Comparison: For a computationally constrained embedded system, lightweight CNNs generally offer an easier path to low-latency inference. Efficient ViTs can be competitive but may require more optimization and hardware resources.

2: Suitability for Real-Time Deployment, Transfer Learning and Data Augmentation

Question: Discuss which method is better suited for real-time deployment and how transfer learning or data augmentation could impact performance.

Answer: For the stated scenario, a lightweight CNN-based object detector is generally the safer choice for real-time embedded deployment because CNNs offer efficient inference, mature optimization tools and strong multi-scale object detection. However, an efficient hybrid or transformer-based detector may be selected if the embedded hardware has sufficient acceleration and the model demonstrates better validation performance.

Real-time deployment: A practical solution is to use a lightweight pretrained CNN detector, optimize it using quantization or pruning, and run it on an embedded GPU, NPU or AI accelerator.

Transfer learning: A model pretrained on a large image dataset can be fine-tuned using the limited labelled obstacle dataset. This allows the model to start with useful visual features instead of learning everything from scratch. Transfer learning is especially important when only a few hundred or few thousand labelled images are available.

Data augmentation: Augmentation can generate additional training variation through horizontal flipping, scaling, cropping, brightness and contrast changes, blur, noise and small rotations. It can help the model become robust to different viewpoints, weather and lighting.

Rare-object augmentation: Synthetic examples generated using generative models can supplement real images of rare obstacles. These samples should be validated so that unrealistic artifacts do not become learned features.

Expected impact: Transfer learning generally improves convergence and performance with limited data, while augmentation reduces overfitting and improves robustness. Together, they can substantially improve detection of rare obstacles.

Overall Comparison for Question 1

Criterion	CNN	Vision Transformer	Preferred for Scenario
Local features	Excellent	Good	CNN
Global context	Good with deep/multi-scale networks	Excellent	ViT
Small-object detection	Strong with multi-scale features	Strong but resolution-sensitive	CNN
Embedded inference	Usually faster/easier to optimize	Often more demanding	CNN
Limited labelled data	Strong with transfer learning	Strong with large-scale pretraining	Depends on pretrained model
Real-time deployment	Very suitable with lightweight models	Possible with efficient variants	CNN

2: PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. The meshes vary in topology and vertex density.

1: Comparison of PCA-Based Shape Priors and Graph-Based Learning

Question: Compare PCA-based shape priors and graph-based learning in terms of ability to handle non-rigid deformations, robustness to varying mesh resolutions and noise, and ease of implementation and computational cost.

A. Ability to Handle Non-Rigid Deformations

PCA-based shape priors – Answer: PCA learns the dominant statistical variations from a training set of aligned shapes. It can represent common deformations such as changes in facial proportions, but it is fundamentally a linear model and may struggle with highly complex or nonlinear deformations.

Graph-based learning – Answer: Graph Neural Networks can learn local geometric relationships and nonlinear feature transformations from mesh connectivity. They are better suited to learning complex deformation patterns when adequate training data are available.

Comparison: Graph-based learning generally provides greater flexibility for non-rigid deformations, while PCA is simpler and effective when shape variations are approximately linear and represented well by the training set.

B. Robustness to Varying Mesh Resolutions and Noise

PCA-based shape priors – Answer: PCA normally requires shapes to be aligned and represented consistently. Different vertex counts or sampling patterns must be handled through correspondence, resampling or a common representation before PCA can be applied.

Graph-based learning – Answer: Graph methods operate directly on irregular mesh structures and can use vertex connectivity, local geometry and learned features. They can therefore be more flexible when meshes have different resolutions.

Noise handling: Both methods require preprocessing for severe noise. Graph models can learn robust local features, while PCA may be sensitive to noisy training shapes because noise can influence the estimated mean and principal components.

Comparison: Graph-based learning is generally more flexible for varying mesh resolution and topology, whereas PCA works best when the meshes have been normalized to a common representation.

C. Ease of Implementation and Computational Cost

PCA-based shape priors – Answer: PCA is relatively easy to implement using standard linear algebra libraries. Once the model is trained, projection and reconstruction are computationally inexpensive.

Graph-based learning – Answer: Graph learning requires graph construction, feature design or learning, model training and potentially substantial GPU memory. Large meshes containing tens of thousands of vertices can be computationally demanding.

Comparison: PCA is simpler, cheaper and easier to interpret. Graph-based learning is more complex and computationally expensive but provides greater modelling flexibility.

2: Best Approach for Accurate Correspondences and Realistic Morphing

Question: Analyze which approach would provide more accurate correspondences for realistic morphing and why.

Answer: For realistic morphing between 3D human faces with varying topology and vertex density, graph-based learning is generally the stronger choice, provided that sufficient training data and computational resources are available.

Reason 1 – Nonlinear facial variation: Human faces can vary through expressions, age-related changes, facial proportions and other non-rigid deformations. Graph-based models can learn nonlinear relationships between local geometry and corresponding facial regions.

Reason 2 – Mesh connectivity: Graph learning uses the relationships between neighboring vertices. This provides local surface information that is important for matching regions such as the nose, eyes, lips and cheeks.

Reason 3 – Different mesh resolutions: Meshes may contain different numbers of vertices. Graph-based representations do not require direct matching of vertex indices and can use learned geometric descriptors instead.

Reason 4 – Better morphing consistency: Accurate correspondences are essential for realistic morphing. Graph-based features can provide smoother and more semantically consistent mappings across the face surface.

Role of PCA: PCA remains useful when the faces are already aligned and have a common representation. It provides a compact statistical model and is computationally inexpensive. It can also serve as a baseline or prior within a larger correspondence pipeline.

Final choice: Graph-based learning is preferred for the stated scenario because the meshes vary in topology and vertex density and realistic facial morphing requires accurate, nonlinear correspondences. PCA is preferred when simplicity, low computational cost and a relatively consistent shape representation are the main requirements.

Overall Comparison for Question 2

Criterion	PCA Shape Prior	Graph-Based Learning	Preferred for Scenario
Non-rigid deformation	Moderate; mainly linear variation	Strong; nonlinear learning	Graph
Varying mesh resolution	Requires common representation	More flexible	Graph
Noise robustness	Moderate	Can learn robust local features	Graph
Implementation	Simple	More complex	PCA
Computational cost	Low	Higher	PCA
Realistic face correspondence	Good for aligned shapes	Potentially more accurate	Graph

Final Conclusion

CNNs are generally more practical for real-time object detection on embedded autonomous-vehicle platforms because of their efficient local feature extraction, multi-scale detection capability and mature optimization methods. Vision Transformers provide powerful global-context modelling and can be competitive when supported by sufficient data and hardware. Transfer learning and data augmentation are important for limited labelled datasets. For 3D face correspondence, PCA is simpler and computationally cheaper, but graph-based learning is more flexible for nonlinear deformations, varying mesh resolution and topology. Therefore, graph-based learning is generally better suited for highly accurate and realistic 3D face morphing when adequate training data and computational resources are available.